The Chinese University of Hong Kong, Shenzhen
SDS · School of Data Science

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MAT 3007 — Optimization

Exercise Sheet 9

Problem 1 (Modeling): (*50* pts)

A company produces three types of products, A, B, and C. Each product requires time on two different machines: Machine 1 and Machine 2. The production requirements, and profits are as follows:

- Each unit of product A requires 3 hours on Machine 1 and 2 hours on Machine 2, with a profit of \$30 per unit.
- Each unit of product B requires 2 hours on Machine 1 and 4 hours on Machine 2, with a profit of \$40 per unit.
- Each unit of product C requires 1 hour on Machine 1 and 3 hours on Machine 2, with a profit of \$50 per unit.

The machines have the following availability:

- Machine 1: 60 hours
- Machine 2: 72 hours

Additionally, the company faces the following cost and constraints:

a) Due to marketing restrictions:
   - At least 8 units of A must be produced.
   - At least 10 units of B must be produced.
   - At least 5 units of C must be produced.

b) Due to storage limitations:
   - If the total number of products produced exceeds 20 units, a fixed storage cost of \$200 will be incurred.
   - The total number of products produced cannot exceed 40 units.

c) The company wants to ensure that no more than 70% of the total production is allocated to any single product.

Formulate an integer optimization problem. Clearly define the decision variables, objective function, and constraints.

Problem 2 (Branch and Bound): (*50* pts)

Solve the following integer programming problem using the Branch and Bound method by hand (you can stop after you find one optimal solution):

$$\begin{aligned} \text{maximize} \quad & 3x_1 + 4x_2 \\ \text{subject to} \quad & 2x_1 + 3x_2 \leq 12 \\ & 3x_1 + 2x_2 \leq 12 \\ & x_1, x_2 \geq 0, \quad x_1, x_2 \in \mathbb{Z} \end{aligned}$$

Hint: For two-dimensional linear programming problems, you can solve them graphically or use any method of your choice.

Sheet 9 is due on **May 14th**. Submit your solutions before **May 14th**, **11:00 pm**.